WORKED SOLUTION

STEP I 2010, Question 4

Pure • Integration by Substitution

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s1-q4

The key idea

The substitution is chosen so that x and $x+1$ both become simple fractions with the same denominator t^2-1, which makes the square root of their product come out exactly. The integral then turns into the standard one in the note. For the volume, squaring y produces two easy logarithms and a cross term, and the cross term is precisely the integral we have just found.

Solution

The integral

Take $x>0$. The substitution $x=\dfrac{1}{t^2-1}$ with $t>1$ is legitimate: solving for t gives

$$t=\sqrt{1+\frac{1}{x}}=\frac{\sqrt{x+1}}{\sqrt{x}},$$

and as t runs over $(1,\infty)$, x runs over $(0,\infty)$ (decreasingly), one value of x for each t.

Now

$$x+1=\frac{1}{t^2-1}+1=\frac{t^2}{t^2-1}, \qquad \text{so} \qquad x(x+1)=\frac{t^2}{(t^2-1)^2}.$$

Since $t>1$, both t and t^2-1 are positive, so

$$\sqrt{x(x+1)}=\frac{t}{t^2-1}.$$

? **Ask yourself** Why did we need $t>1$ when taking the square root?

A square root is never negative, so $\sqrt{t^2/(t^2-1)^2}=|t|/|t^2-1|$. The condition $t>1$ lets us drop both modulus signs. It also explains the choice: $t^2-1>0$ is exactly what makes $x>0$.

Differentiating, $\dfrac{\mathrm{d}x}{\mathrm{d}t} = -\dfrac{2t}{(t^2-1)^2}$. Therefore

$$\int \frac{1}{\sqrt{x(x+1)}}\,\mathrm{d}x = \int \frac{t^2-1}{t}\cdot\left(-\frac{2t}{(t^2-1)^2}\right)\mathrm{d}t = -2\int \frac{1}{t^2-1}\,\mathrm{d}t$$
$$= -2\cdot\frac{1}{2}\ln\left|\frac{t-1}{t+1}\right| + c = \ln\left(\frac{t+1}{t-1}\right) + c,$$

using the note with $a = 1$, and the fact that $t > 1$ makes $(t-1)/(t+1)$ positive.

Now return to x using $t = \sqrt{x+1}/\sqrt{x}$. Multiplying top and bottom by $\sqrt{x}$,

$$\frac{t+1}{t-1} = \frac{\sqrt{x+1}+\sqrt{x}}{\sqrt{x+1}-\sqrt{x}} = \frac{\left(\sqrt{x+1}+\sqrt{x}\right)^2}{(x+1)-x} = \left(\sqrt{x}+\sqrt{x+1}\right)^2,$$

where we multiplied top and bottom by $\sqrt{x+1}+\sqrt{x}$ and used the difference of two squares. Hence

$$\int \frac{1}{\sqrt{x(x+1)}}\,\mathrm{d}x = 2\ln\left(\sqrt{x}+\sqrt{x+1}\right) + c,$$

as required.

The volume

The curve has $y > 0$ for $x > 0$, since $\sqrt{x} < \sqrt{x+1}$ gives $1/\sqrt{x} > 1/\sqrt{x+1}$. So the volume swept out by the region between the curve and the x-axis is $V = \pi\int y^2\,\mathrm{d}x$. Expanding,

$$y^2 = \frac{1}{x} + \frac{1}{x+1} - \frac{2}{\sqrt{x}\sqrt{x+1}} = \frac{1}{x} + \frac{1}{x+1} - \frac{2}{\sqrt{x(x+1)}},$$

and the last term is twice the integrand from the first part. So

$$V = \pi\Big[\ln x + \ln(x+1)\Big]_{1/8}^{9/16} - 2\pi\Big[2\ln\left(\sqrt{x}+\sqrt{x+1}\right)\Big]_{1/8}^{9/16}.$$

The limits are chosen so that the square roots are tidy:

- at $x = \frac{9}{16}$: $\sqrt{x} = \frac{3}{4}$, $\sqrt{x+1} = \frac{5}{4}$, sum 2, and $x(x+1) = \frac{225}{256}$;
- at $x = \frac{1}{8}$: $\sqrt{x} = \frac{1}{2\sqrt{2}}$, $\sqrt{x+1} = \frac{3}{2\sqrt{2}}$, sum $\frac{4}{2\sqrt{2}} = \sqrt{2}$, and $x(x+1) = \frac{9}{64}$.

The first bracket is

$$\ln\frac{225}{256} - \ln\frac{9}{64} = \ln\left(\frac{225}{256}\cdot\frac{64}{9}\right) = \ln\frac{25}{4},$$

and the second is

$$2\ln 2 - 2\ln\sqrt{2} = 2\ln 2 - \ln 2 = \ln 2.$$

Therefore

$$V = \pi\ln\frac{25}{4} - 2\pi\ln 2 = \pi\ln\frac{25}{4} - \pi\ln 4 = \pi\ln\frac{25}{16} = 2\pi\ln\frac{5}{4},$$

which is the result we wanted.

Checks

- Differentiate the answer:

$$\frac{\mathrm{d}}{\mathrm{d}x}\,2\ln\bigl(\sqrt{x}+\sqrt{x+1}\bigr) = \frac{2}{\sqrt{x}+\sqrt{x+1}}\left(\frac{1}{2\sqrt{x}}+\frac{1}{2\sqrt{x+1}}\right) = \frac{1}{\sqrt{x}\sqrt{x+1}},$$

  since the bracket equals $\dfrac{\sqrt{x+1}+\sqrt{x}}{2\sqrt{x}\sqrt{x+1}}$.

- Numerically, Simpson's rule gives $\pi\int_{1/8}^{9/16} y^2\,\mathrm{d}x \approx 1.402052$, matching $2\pi\ln\frac{5}{4} \approx 1.402052$. Simpson's rule also confirms $\int_{1/8}^{9/16} \frac{\mathrm{d}x}{\sqrt{x(x+1)}} = \ln 2 \approx 0.693147$.

Common slips

- Forgetting the minus sign in $\mathrm{d}x/\mathrm{d}t$. It is what turns $\ln\frac{t-1}{t+1}$ into $\ln\frac{t+1}{t-1}$.
- Expanding y^2 as $\frac{1}{x}-\frac{1}{x+1}$: the cross term $-2/\sqrt{x(x+1)}$ is the whole point of the first part.
- Forgetting the factor π, or using 2π instead of π, in the volume formula.

Beyond STEP

The answer $2\ln(\sqrt{x}+\sqrt{x+1})$ is really $2\operatorname{arsinh}\sqrt{x}$, because $\operatorname{arsinh} u = \ln\bigl(u+\sqrt{u^2+1}\bigr)$. The substitution $x=\sinh^2\theta$ gives the same result in one line. Choosing the substitution that rationalises a square root is a systematic technique (the Euler substitutions), and it lies behind the result that every integral of a rational function of x and $\sqrt{ax^2+bx+c}$ can be done in elementary functions. Siklos, *Advanced Problems in Mathematics*, has many more integrals in this spirit.

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